MINI PROJECT REPORT

ON

ELECTRICITY CONSUMPTION ANALYSIS SYSTEM

Using Python, Pandas and Matplotlib

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Abstract

The Electricity Consumption Analysis System is a Python-based mini-project designed to analyse monthly electricity usage. The application reads data from a CSV file, validates the records, calculates total and average consumption, identifies the months with the highest and lowest usage, estimates bills using a configurable example rate, and displays results through charts. Pandas is used for data processing and Matplotlib is used for visualization. The project demonstrates how programming can help users understand consumption patterns and make informed energy-saving decisions. The included dataset is sample data and should be replaced with actual readings for real analysis.

1. Introduction

Electricity is an essential resource in homes, colleges, offices and industries. Tracking electricity consumption can help users understand usage patterns, notice unusually high consumption and plan ways to save energy. Analysing readings manually can be time-consuming, especially across several months. This project automates basic calculations and presents the results using clear graphs.

2. Problem Statement

Users may have monthly electricity readings but lack a simple method to compare consumption, calculate an estimated energy charge and view trends. The proposed system processes a structured dataset and presents a summary and visual reports.

3. Objectives

- Store monthly electricity consumption in a CSV file.
- Read, clean and sort data using Python and Pandas.
- Calculate total and average consumption.
- Identify the months with the highest and lowest consumption.
- Estimate monthly energy charges using a configurable rate.
- Generate bar charts and a line chart for easy comparison.
- Export the processed analysis to a CSV file.

4. Scope of the Project

The project is intended for academic demonstration and small datasets. It analyses monthly readings provided by the user. It does not connect to an actual meter, predict future demand, or reproduce a utility provider's full billing rules.

5. Hardware and Software Requirements

Requirement	Specification
Hardware	Computer or laptop; 4 GB RAM recommended
Operating system	Windows, Linux or macOS
Language	Python 3.10 or later recommended
Libraries	Pandas and Matplotlib
Input format	CSV file
Output	Terminal summary, PNG charts and CSV report

6. System Design / Workflow

1. Start the program.
2. Load monthly records from data/electricity_data.csv.
3. Validate the required columns and numeric consumption values.
4. Sort records by month and calculate the estimated bill.
5. Display summary statistics in the terminal.
6. Create consumption and bill charts.
7. Export the processed data to the output folder.
8. End the program.

7. Dataset Description

The supplied sample file contains one record for each month of 2026. The Month column uses the YYYY-MM format and Units_kWh records electricity consumption in kilowatt-hours. The values are illustrative and are not actual household readings.

Month	Units (kWh)
2026-01	120
2026-02	150
2026-03	180
2026-04	140
2026-05	210
2026-06	195

8. Methodology

Pandas reads the CSV file and converts the Month column into dates. Invalid dates and non-numeric consumption entries are removed, as are negative readings. Duplicate months are reduced to the last record for that month. The system then calculates total consumption, average consumption, maximum and minimum consumption, and an estimated bill for each month.

Estimated energy charge = Units consumed (kWh) × Assumed rate per kWh

The default example rate is ₹6 per kWh. This is configurable and is not intended to represent every electricity provider's tariff. Actual bills may include fixed charges, slab rates, taxes and other fees.

9. Implementation

The main program is electricity_analysis.py. The load_data() function reads and validates the dataset. print_summary() displays the key figures. create_charts() generates the monthly consumption bar chart, consumption trend line chart and estimated bill chart. export_analysis() saves the processed results.

The complete source code is supplied separately in the project ZIP. The program uses relative paths, so it can be run from the extracted project folder without editing file locations.

10. How to Run the Project

9. Install Python 3.10 or later.
10. Extract the project ZIP file.
11. Open a terminal or command prompt in the extracted folder.
12. Run: python -m pip install -r requirements.txt
13. Run: python electricity_analysis.py
14. Review the terminal summary, chart windows and generated files in the output folder.

On some Windows systems, use the command py instead of python.

11. Expected Results

The program displays the number of months analysed, total consumption, average monthly consumption, estimated total energy charge, and the highest- and lowest-consumption months. It then displays three charts and saves them as PNG files. A processed CSV file containing monthly usage and estimated bills is also created.

With the included sample dataset, the highest usage is 230 kWh in July 2026 and the lowest usage is 120 kWh in January 2026. The total sample consumption is 2,175 kWh. At the example rate of ₹6 per kWh, the estimated total energy charge is ₹13,050.

12. Testing

Test case	Expected behaviour	Status
Valid CSV	Summary and charts are produced	Pass
Missing dataset	Clear error message is displayed	Pass
Invalid or blank rows	Invalid rows are ignored	Pass
Negative consumption	Negative rows are ignored	Pass
Change unit rate	Estimated bills update	Pass

accordingly

13. Advantages

- Simple and low-cost to use.
- Automates repetitive calculations.
- Charts make month-to-month comparisons easier.
- CSV input can be edited in a spreadsheet application.
- Can be extended with a graphical interface or a database.

14. Limitations

- The supplied data is sample data.
- The estimated bill uses a single flat rate.
- The program depends on correctly formatted CSV input.
- It does not connect to smart meters or an electricity provider.

15. Future Enhancements

- Add a Tkinter graphical user interface.
- Store readings in an SQLite database.
- Support daily readings and multiple households.
- Use actual tariff slabs and fixed charges.
- Add consumption forecasting and energy-saving recommendations.
- Create a web dashboard for remote access.

16. Conclusion

The Electricity Consumption Analysis System demonstrates how Python, Pandas and Matplotlib can be used to process electricity data and present useful insights. It calculates consumption statistics, estimates energy charges and creates visual comparisons. The project meets its learning objectives and provides a foundation for more advanced energy-monitoring applications.

17. References

- Python documentation: https://docs.python.org/3/
- Pandas documentation: https://pandas.pydata.org/docs/
- Matplotlib documentation: https://matplotlib.org/stable/